

Roopesh Bharatwaj KR

AI Architect | Applied AI Lead | Principal AI Engineer

Ireland | [Gmail](#) | [LinkedIn](#) | [Github](#)

PROFESSIONAL PROFILE

AI architect and software engineering leader with 11+ years of experience across applied research, product development and technical consulting. Designs and delivers LLM, retrieval-augmented generation, computer vision and MLOps solutions, while remaining hands-on with Python, APIs and cloud deployment. At CeADAR and the NovaUCD AI Accelerator, works directly with founders, executives and engineering teams to turn business requirements into practical architectures, delivery plans and production-ready systems.

PROFESSIONAL EXPERIENCE

Senior Data Scientist and AI Technical Lead

Jan 2024 – Present

CeADAR

Ireland

- Own end-to-end delivery of concurrent AI engagements for SMEs, public-sector organisations and startups, from client discovery and requirements definition through proof of concept, production hardening, cloud deployment and handover.
- Designed an AI-assisted financial-document extraction workflow that replaced a benchmark task requiring approximately three days of manual analysis; the automated result was within approximately 4% of the manually prepared financial model and also exposed an error in the source material.
- Architected a multi-agent examination assessment and feedback system for an education provider, agentic RAG applications that reduced manual information-retrieval workflows, and a multilingual conversational AI platform serving public-sector users across multiple counties.
- Architected and delivered an immersive generative-AI platform combining Stability AI, fine-tuned LoRA models and live weather APIs to produce dynamic digital interpretations of Irish landmarks and traditional oil paintings.
- Own cloud governance and FinOps across AWS, Azure and GCP, including organisational setup, billing oversight, cost monitoring, VAT coordination, research-credit applications and provider support.
- Define solution architecture, security, evaluation, observability and deployment standards; review code and designs, mentor engineers and interns, and coordinate technical delivery across a team of more than 10 data scientists.

Lead Technical Consultant, NovaUCD AI Accelerator

2024 – Present

CeADAR and NovaUCD

Ireland

- Provided technical consulting to more than 30 early-stage companies over two years on AI adoption, system design, cloud architecture, model selection, evaluation and product delivery.
- Conduct architecture, technical-feasibility and product reviews, helping founders connect market requirements with maintainable system design, data flows, APIs, security, observability and cloud cost.
- Perform authorised manual and tool-assisted penetration testing of startup web applications and APIs, document reproducible findings and business impact, and validate remediation across authentication, access control, tenant isolation and data-exposure risks.
- Advise startups on responsible AI adoption and the EU AI Act, including system classification, intended use, data governance, transparency, human oversight and risk-management controls.

Data Scientist II

Feb 2023 – Dec 2023

CeADAR

Ireland

- Led research and development on graph machine learning, including tabular-to-graph conversion in Neo4j, graph convolutional networks, link prediction and node classification for real-time client data.
- Managed client communication and project delivery and led a five-person team spanning data science, software engineering and research internships.

EARLIER EXPERIENCE

Research ML Engineer

Sep 2021 – Feb 2023

COMMAND Gateway Technologies: SRI-TUS

Ireland

- Owned AI delivery across client projects for NPD Group, Enterprise Ireland and Luxottica, covering requirements, architecture, model development, APIs, testing and cloud deployment.
- Designed and operated Azure data and ML pipelines processing approximately 25–30 million consumer transaction and receipt records each week, including data from NPD's ReceiptPal application, with ingestion, validation, cleaning and feature engineering in Azure Data Lake.

- Orchestrated more than 5,000 anomaly-detection model runs per processing cycle using Azure Machine Learning, including automated retraining, drift monitoring, Feast-based feature management and deployment automation.
- Developed NLP systems for text classification and semantic search using transformer models, and computer-vision components for OCR, facial analysis and product recommendation.
- Built iris-colour and face-shape classification models for a Ray-Ban recommendation application and deployed models to mobile and edge environments.
- Developed perception pipelines for a robotic arm using object detection, depth estimation, semantic segmentation, OpenVINO and OAK-D devices.

AI Researcher

DilSeBol Pvt Ltd

Apr 2020 – May 2021

Remote, India and Ireland

- Researched and developed computer-vision systems for vehicle and licence-plate detection, including YOLO-based model tuning, data augmentation, edge inference and containerised deployment.

Full Stack Developer and ML Engineer

AdmitStory Limited

Nov 2018 – Mar 2020

India and United Kingdom

- Managed on-site and offshore software and ML teams, served as Scrum Master and reviewed code for an eight-person development team.
- Built NLP-based admission-ranking, conversational AI and document-processing applications for universities, including REST APIs and web integration.

AI Software Developer

DilSeBol Pvt Ltd

Jan 2015 – Sep 2018

India

- Developed OCR, NLP, forecasting and web applications for medical-document processing, insurance analytics and customer-feedback analysis; also managed client delivery and project planning.
- Received the company's Best Employee award across three consecutive review periods in 2017 and 2018.

EDUCATION

Technological University of the Shannon TUS

MS in Software Design with Artificial Intelligence

Ireland

2020 – 2021

Anna University

B.Tech in Information Technology

India

2011 – 2015

SELECTED PUBLICATIONS AND RECOGNITION

“EHRCoI4: A Novel Framework for Enhancing Human-Robot Collaboration in Industry 4.0,” IEEE eSmarTA 2023.

Top 50 worldwide finalist, OpenCV Spatial AI Contest 2021, organised with Microsoft Azure and Intel.

“Effective Scheduling and Allocation of Virtual Machines in Cloud Computing,” Anna University, 2015; received the conference award for meritorious paper performance.

TECHNICAL EXPERTISE

AI Architecture: LLMs, RAG, multimodal AI, model evaluation, multi-agent workflows, speech AI, computer vision

Engineering: Python, Java, R, FastAPI, React, TypeScript, REST APIs, SQL, Git, Docker, CI/CD

Cloud and MLOps: AWS, Azure, GCP, SageMaker, Bedrock, Vertex AI, Azure ML, Kubernetes, model monitoring

Leadership: Solution architecture, technical discovery, stakeholder management, code review, mentoring, delivery planning